Admission Control for LLM Inference Serving:
A Systematic Review

Submission to ACM Computing Surveys 26 September 2026

Dear Editors,

We are submitting "Admission Control for LLM Inference Serving: A Systematic Review" for consideration at ACM Computing Surveys.

The paper reviews fifty-five works published between 2022 and 2026 on the decision made before a request enters an LLM serving system: admit it, hold it, or refuse it. We followed an adapted PRISMA protocol, and the per-paper coding sheets are released with the paper so the classification can be checked.

CSUR describes a survey article as one that emphasises classification of existing literature, developing a perspective, and evaluating trends. That is what we set out to do. The taxonomy classifies the corpus on five axes.

Three findings came out of it.

Admission control turns out to sit at a layer the serving literature does not study. We took the nine serving systems that the admission-primary papers most often build on or compare against, and all nine handle overload inside the engine through batching, migration, placement, autoscaling or memory management. None takes a decision at the ingress. The corpus work that does gate at ingress is a control-plane contribution rather than an engine one, which is the distinction that matters here; it is not about venue, and we say in the paper that we would draw the same conclusion had that work appeared alongside the nine.

We are careful about how this reads. Those nine are scheduling and memory-management papers, and refusing work at the door is outside the problem each of them set for itself. It is an observation about where attention has gone, not a criticism of those systems.

No two of the sixteen admission-primary papers share a baseline set. Policies are measured against the absence of admission control and almost never against one another, so a practitioner cannot use the literature to decide which policy suits their deployment.

The third finding came out of testing whether the taxonomy was usable. We placed all sixteen papers on it, recording seven properties each: the five axes, plus what a policy assumes an arriving request carries, plus the range its signal took during the paper's own evaluation. Six of the seven filled from the published text, and all five axes were among them. One property did not fill for any paper. None reports the range its own signal took during the evaluation, which means a policy whose threshold was never approached and a policy that evaluated a varying signal and declined to reject produce the same reported numbers. We state this as an absence in our extraction rather than as a proof that no such reporting exists anywhere, and we have written a threats-to-validity section that says so plainly.

On positioning: five surveys of LLM inference serving precede this one. Zhen et al., Heisler et al., and Li et al. cover efficient serving broadly and treat admission as one component among many. Mitzenmacher and Shahout give the queueing-theoretic view and enumerate a theoretical agenda; our open problems overlap theirs on reasoning workloads and on schedulability, and we say so in Section 10 rather than claiming the lists are disjoint. Ma et al. address load balancing, which is routing rather than ingress. None of the five makes admission the primary axis, and none audits what the literature reports.

Two disclosures. First, the screening and coding were done by one reader, so there is no inter-rater agreement statistic; we have owned that in the threats-to-validity section and released the coding sheets so the placements can be disputed. Second, we intend to post a preprint to arXiv. If it is up before review concludes we will note the prior appearance on the title page, and we understand this does not count as prior publication under CSUR's policy. We have not submitted this work anywhere else.

The manuscript runs to 30 pages in the acmsmall template, which we understand falls under the full survey category rather than the short one.

Thank you for considering it.

Sincerely,

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